Quality Management and Statistical Process Control

Statistical Process Control (SPC) & Process Capability



MGT 207 — Production and Operations Management

Natural Manufacturing Variations



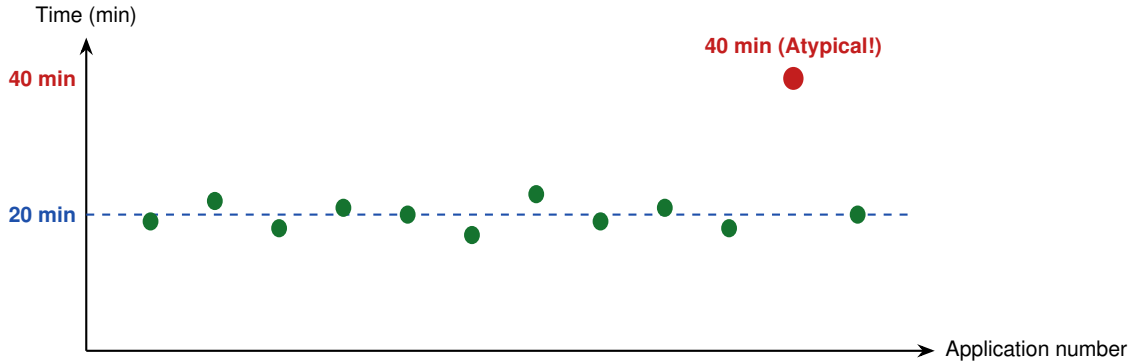
Every Process Exhibits Variation

Even highly automated bottling lines exhibit small, natural (random) variations in fill height from bottle to bottle.

All Processes Have Some Variation. . .

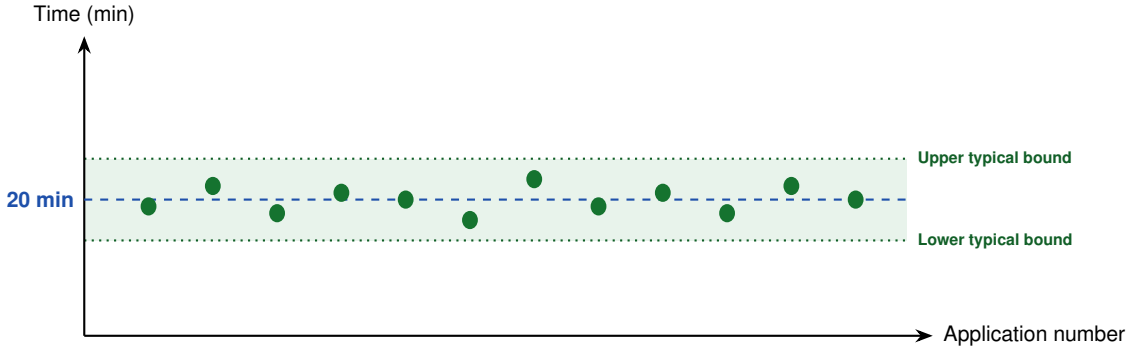
Typical Variation vs. Atypical Variation

Example: The time taken to rate an insurance policy averages **20 minutes**. What would you think if a particular policy took **40 minutes**?



Process is In-Control if Measurements Fall Within the Typical Range

This process is staying in control.

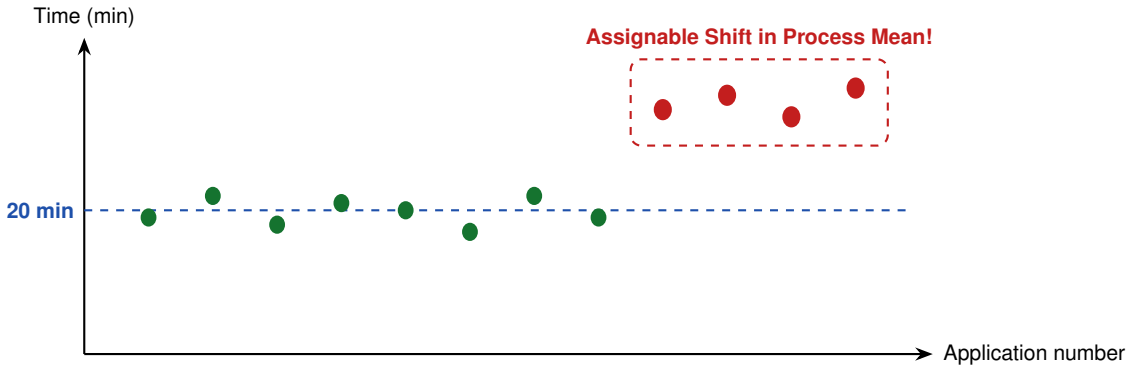


Important Distinction

In-control does not necessarily mean high quality—it just means the process is behaving consistently as you expect it to behave (only natural/common-cause variation is present).

A Change In Measurements Indicates Process May Be Out-of-Control

A significant change indicates that the process is no longer behaving as you expect it to behave.



Quality Testing by Sampling

Goal of Statistical Process Control

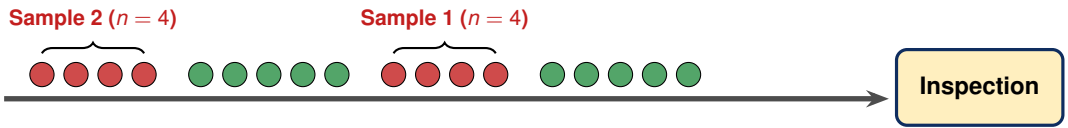
Detect whether there is:

1. A **shift in the process mean**, or
2. An **increase in process variation**.

Sampling Protocol

A **sample** is a set of measurements taken periodically:

- **Sample frequency:** e.g., every 5 products
- **Sample size (n):** e.g., $n = 4$ units per sample



Some Examples of Sample Data. . .

Sample	Measurements ($n = 4$)				Mean ($\bar{X}$)	Range ($R = \max - \min$)
1	97.06	97.32	100.19	101.02	**98.90**	**3.96**
2	97.68	100.05	97.80	97.85	**98.34**	**2.37**
3	98.76	98.22	100.72	103.11	**100.20**	**4.89**
4	101.34	100.00	103.19	105.79	**102.58**	**5.79**
5	102.78	105.28	105.02	97.96	**102.76**	**7.32**
6	104.23	104.57	97.50	100.39	**101.67**	**7.07**

Sample Mean ($\bar{X}_i$)

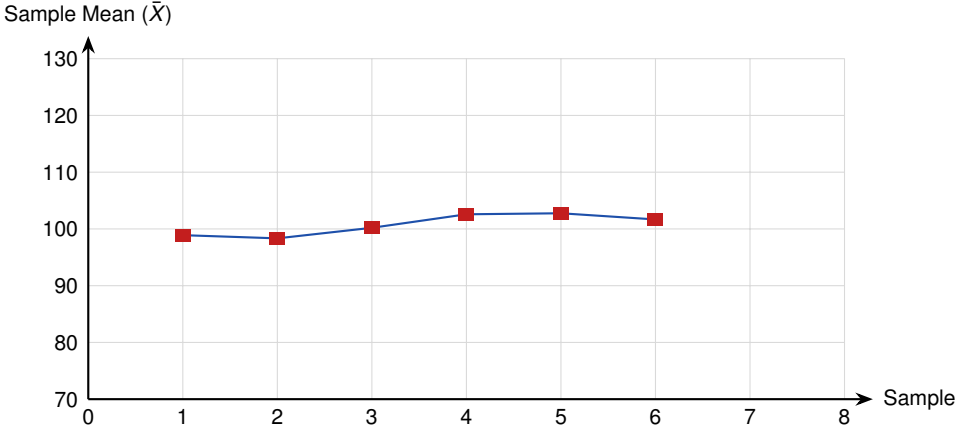
Measures the central tendency of sample i:
$\bar{X}_1 = \frac{97.06+97.32+100.19+101.02}{4} = \mathbf{98.90}$.

Sample Range (R_i)

Measures dispersion within sample i:
$R_1 = 101.02 - 97.06 = \mathbf{3.96}$.

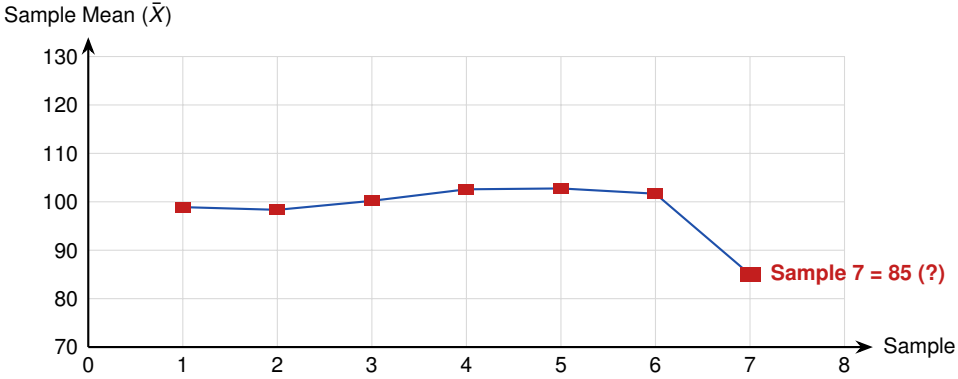
Visualizing the Data Can Be Helpful

A plot of sample means ($\bar{X}_1 \ldots \bar{X}_6$)



Let's Look at the Next Sample Mean

Say Sample 7 comes in with a mean of 85:

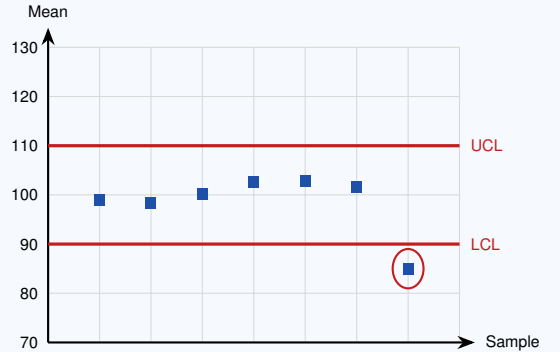


Key Question

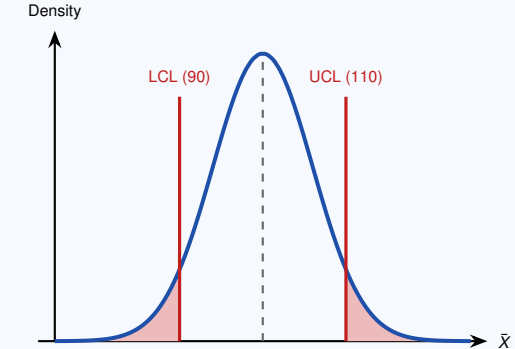
How might you decide if 85 is atypical for this process?

One Approach Is to Specify Limits on What Is Typical

Anything Outside the Limits Is Abnormal

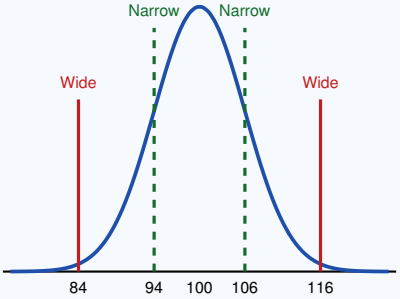


Sample Means Are Normally Distributed



But What Are the Right Limits? Wide or Narrow?

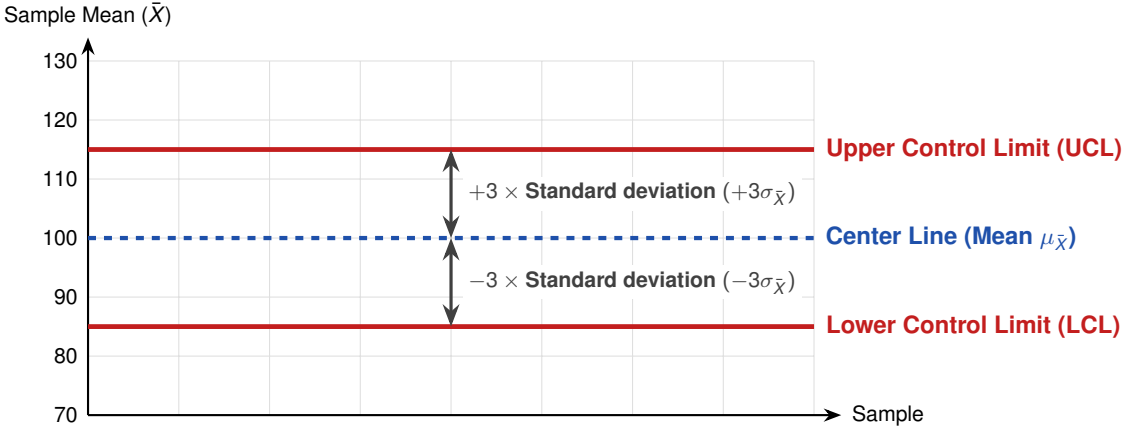
Wide vs. Narrow Limits



Pros and Cons of Limit Width

- **Narrow Control Limits** (e.g., $\pm 1\sigma$ or $\pm 2\sigma$):
  - **Pro:** Quickly detects small process shifts (low Type II error).
  - **Con:** Frequent **false alarms** when the process is actually in control (high Type I error).
- **Wide Control Limits** (e.g., $\pm 3\sigma$):
  - **Pro:** Very few false alarms (99.73% of normal sample means fall inside $\pm 3\sigma$).
  - **Con:** Slower to detect small/moderate shifts.

Often the Limits Are Set As Mean ± 3 Standard Deviations



How Many Defects Per Million?

Mathematical Representation for $\pm 3\sigma$ Limits

1. **Calculate Cumulative Normal Probabilities (Z-Scores):**
   - For $+3\sigma$: $\Phi(+3) =$ `=NORM.S.DIST(3, TRUE)` $=$ **0.99865**
   - For -3σ: $\Phi(-3) =$ `=NORM.S.DIST(-3, TRUE)` $=$ **0.00135**
2. **Calculate Probability Outside $\pm 3\sigma$ Limits:**

$$P(|Z| > 3) = (1 - 0.99865) + 0.00135 = \mathbf{0.0027}\ (\mathbf{0.27}\%)$$

3. **Convert to Defects Per Million Opportunities (DPMO):**

$$\text{Defects per Million} = 10^6 \times 0.0027 = \mathbf{2{,}700} \text{ per million}$$

What Mean and Standard Deviation Should We Use?

1. Sample Means' Mean ($\mu_{\bar{X}}$)

We estimate the process mean using the **mean of the sample means** ($\bar{\bar{X}}$):

$$\mu_{\bar{X}} = \bar{\bar{X}} = \frac{1}{N}\sum_{i=1}^{N} \bar{X}_i$$

- N = number of samples
- $\bar{X}_i$ = sample mean for the ith sample

2. Sample Means' Std. Dev. ($\sigma_{\bar{X}}$)

By the Central Limit Theorem, the standard deviation of the sample mean is:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

- σ = process standard deviation
- n = sample size (measurements per sample)

Calculating Process Standard Deviation

Estimating σ from Sample Ranges ($\bar{R}$)

In practice, the true process standard deviation σ is estimated from the **average of the sample ranges** ($\bar{R}$):

$$\sigma = \frac{\bar{R}}{d_2} \qquad \text{where} \qquad \bar{R} = \frac{\sum_{i=1}^{N} R_i}{N}$$

- R_i = range of sample i ($\max - \min$)
- $\bar{R}$ = average of the sample ranges across N samples
- d_2 = statistical constant found in tables that depends only on sample size n

Bringing It All Together: $\bar{X}$-Chart Control Limits

Deriving the A_2 Formula for $\bar{X}$-Chart Limits

Combining $\sigma_{\bar{X}} = \dfrac{\sigma}{\sqrt{n}}$ and $\sigma = \dfrac{\bar{R}}{d_2}$ with $A_2 = \dfrac{3}{\sqrt{n}\,d_2}$:

$$\mathbf{UCL_{\bar{X}}} = \mu_{\bar{X}} + 3\sigma_{\bar{X}} = \bar{\bar{X}} + \frac{3\bar{R}}{\sqrt{n}\,d_2} = \bar{\bar{\mathbf{X}}} + \mathbf{A_2}\bar{\mathbf{R}}$$

$$\mathbf{LCL_{\bar{X}}} = \mu_{\bar{X}} - 3\sigma_{\bar{X}} = \bar{\bar{X}} - \frac{3\bar{R}}{\sqrt{n}\,d_2} = \bar{\bar{\mathbf{X}}} - \mathbf{A_2}\bar{\mathbf{R}}$$

Master $\bar{X}$-Chart Rule

Control Limits = **Mean of Sample Means** $(\bar{\bar{X}})$ $\pm$ **Constant** (A_2) $\times$ **Mean of Sample Ranges** $(\bar{R})$

Table for A_2 Constants

Critical Caution on Sample Size (n)

Sample size (n) is the number of measurements **in each sample**—and **NOT** the number of samples (N)!

Example Lookup

If you take $N = 6$ samples and each sample contains $n = 5$ measurements, look up $\mathbf{n = 5} \implies \mathbf{A_2 = 0.58}$.

Sample Size (n)	Factor A_2
2	1.88
3	1.02
4	0.73
5	**0.58**
6	0.48
7	0.42
8	0.37
9	0.34
10	0.31
15	0.22
20	0.18

Calculate the Center Line, UCL and LCL for the Following Data

Sample	Measurements ($n = 5$)					Mean ($\bar{X}$)	Range (R)
1	49.59	53.94	47.11	60.58	43.48	50.94	17.10
2	60.41	48.90	29.19	58.87	39.80	47.44	31.22
3	65.89	56.96	44.15	60.76	50.39	55.63	21.74
4	48.48	51.02	57.36	45.54	63.11	53.10	17.56
5	49.10	40.69	42.10	29.26	46.40	41.51	19.84
6	42.76	35.11	47.66	46.78	44.93	**43.45**	**12.55**
Average across $N = 6$ samples:						$\bar{\bar{\mathbf{X}}} = \mathbf{48.68}$	$\bar{\mathbf{R}} = \mathbf{20.00}$

Note: Here $N = 6$ samples, each of sample size $n = 5$ measurements ($\implies A_2 = 0.58$).

Center Line, UCL and LCL ($\bar{X}$-Chart)

Step-by-Step $\bar{X}$-Chart Limits ($n = 5 \implies A_2 = 0.58$)

- **Center Line ($\bar{\bar{X}}$):**

$$\text{Center Line} = \bar{\bar{X}} = \frac{50.94 + 47.44 + 55.63 + 53.10 + 41.51 + 43.45}{6} = \mathbf{48.68}$$

- **Upper Control Limit ($UCL_{\bar{X}}$):**

$$UCL = \bar{\bar{X}} + A_2\bar{R} = 48.68 + (0.58)(20.00) = 48.68 + 11.60 = \mathbf{60.28}$$

- **Lower Control Limit ($LCL_{\bar{X}}$):**

$$LCL = \bar{\bar{X}} - A_2\bar{R} = 48.68 - (0.58)(20.00) = 48.68 - 11.60 = \mathbf{37.08}$$

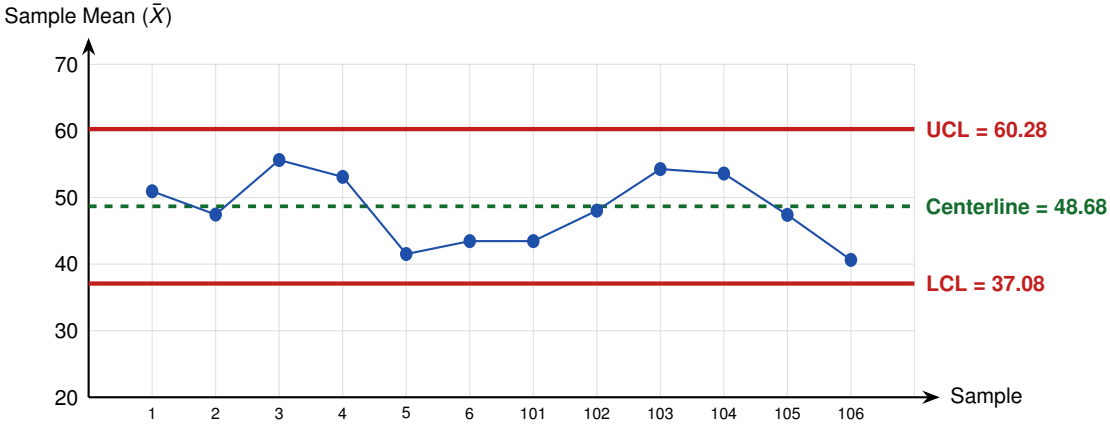
Consider the Following Samples: Is the Process In-Control?

Sample	Measurements ($n = 5$)					Mean ($\bar{X}$)
101	36.62	50.01	44.39	35.84	50.41	**43.45**
102	42.91	34.72	54.82	51.67	56.01	**48.03**
103	57.50	49.69	69.49	43.18	51.50	**54.27**
104	58.36	53.28	55.59	59.06	41.70	**53.60**
105	30.22	16.00	51.87	44.67	94.27	**47.40**
106	2.92	28.45	76.67	66.05	29.04	**40.63**

$\bar{X}$-Chart Evaluation ($LCL = 37.08,\ UCL = 60.28$)

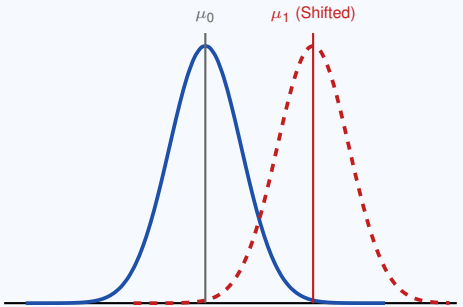
All six sample means fall comfortably inside [37.08, 60.28], so the **process mean has not shifted**. *(But look closely at the individual measurements in Samples 105 and 106!)*

Plotting the Sample Means ($\bar{X}$-Chart)



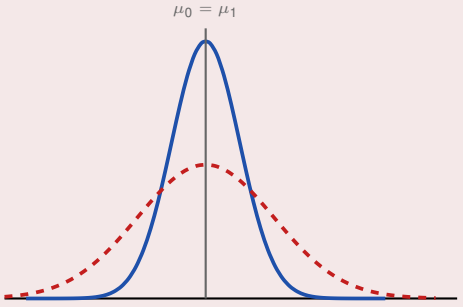
$\bar{X}$-Charts Capture Shift in Means But Not Shift in Variability

$\bar{X}$-Chart Captures Mean Shift



Sample means shift above $UCL_{\bar{X}}$ and trigger an immediate alarm.

$\bar{X}$-Chart Misses Dispersion Spike!

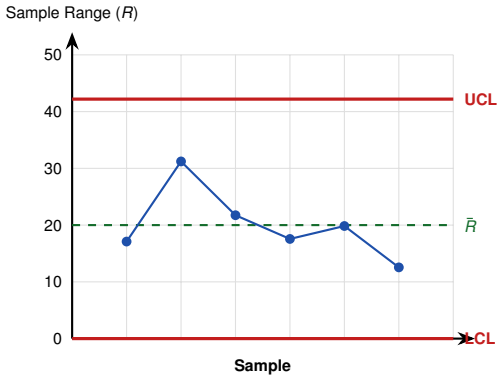


Extreme highs and lows cancel out inside $\bar{X}$, keeping the sample mean near $\bar{\bar{X}}$!

Measuring Shifts in Variability

Range Charts (R-Charts)

Range Charts (R-Charts) Capture Shifts in Variability



How Do We Set R-Chart Limits?

1. **Calculate the Mean Range ($\bar{R}$):**

$$\bar{R} = \frac{\sum_{i=1}^{N} R_i}{N}$$

2. **Calculate Control Limits ($UCL_R,\ LCL_R$):**

$$UCL_R = D_4 \bar{R}$$

$$LCL_R = D_3 \bar{R}$$

(D_3 and D_4 are statistical constants indexed by sample size n.)

Table for D_3 and D_4 Constants

Using D_3 and D_4

- Look up the row corresponding to **sample size** n (number of observations per sample).
- For $n \leq 6$, $D_3 = 0.00$, so $LCL_R = 0$.
- For our 6-sample example with $\mathbf{n = 5}$:

$$D_3 = \mathbf{0.00}, \qquad D_4 = \mathbf{2.11}$$

Sample Size (n)	Factor D_3 (LCL_R)	Factor D_4 (UCL_R)
2	0.00	3.27
3	0.00	2.57
4	0.00	2.28
5	**0.00**	**2.11**
6	0.00	2.00
7	0.08	1.92
8	0.14	1.86
9	0.18	1.82
10	0.22	1.78
15	0.35	1.65
20	0.41	1.59

What Are the R-Chart Control Limits for This Process?

Sample	Measurements ($n = 5$)					Mean ($\bar{X}$)	Range (R)
1	49.59	53.94	47.11	60.58	43.48	50.94	**17.10**
2	60.41	48.90	29.19	58.87	39.80	47.44	**31.22**
3	65.89	56.96	44.15	60.76	50.39	55.63	**21.74**
4	48.48	51.02	57.36	45.54	63.11	53.10	**17.56**
5	49.10	40.69	42.10	29.26	46.40	41.51	**19.84**
6	42.76	35.11	47.66	46.78	44.93	43.45	**12.55**

Use the 6 sample ranges ($n = 5 \implies D_3 = 0.00,\ D_4 = 2.11$) to compute $\bar{R}$, UCL_R, and LCL_R.

Center Line, UCL and LCL Calculations (R-Chart)

Step-by-Step R-Chart Limits ($n = 5 \implies D_3 = 0.00,\ D_4 = 2.11$)

- **Center Line ($\bar{R}$):**

$$\bar{R} = \frac{17.10 + 31.22 + 21.74 + 17.56 + 19.84 + 12.55}{6} = \mathbf{20.00}$$

- **Upper Control Limit (UCL_R):**

$$UCL_R = D_4\bar{R} = 2.11 \times 20.00 = \mathbf{42.20}$$

- **Lower Control Limit (LCL_R):**

$$LCL_R = D_3\bar{R} = 0.00 \times 20.00 = \mathbf{0.00}$$

Consider the Following Samples: Is the Process In-Control?

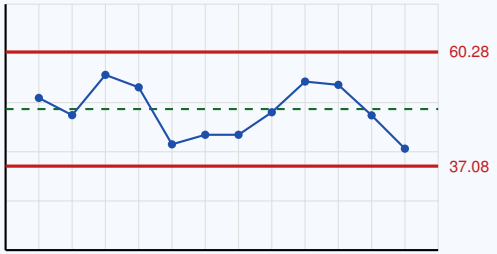
Sample	Measurements ($n = 5$)					Mean ($\bar{X}$)	Range (R)
101	36.62	50.01	44.39	35.84	50.41	43.45	14.58
102	42.91	34.72	54.82	51.67	56.01	48.03	21.28
103	57.50	49.69	69.49	43.18	51.50	54.27	26.31
104	58.36	53.28	55.59	59.06	41.70	53.60	17.36
105	30.22	16.00	51.87	44.67	94.27	47.40	**78.28**
106	2.92	28.45	76.67	66.05	29.04	40.63	**73.75**

R-Chart Verdict ($UCL_R = 42.20$)

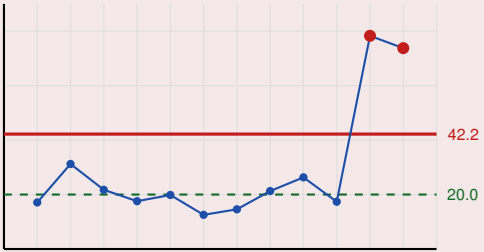
Samples **105** ($R = 78.28$) and **106** ($R = 73.75$) violently breach $UCL_R = 42.20$! The process dispersion is **out of control**.

Out of Control? Need to Look at Both $\bar{X}$-Chart and R-Chart

$\bar{X}$-Chart Looks In Control



R-Chart Reveals Process Is Out of Control!



Discussion Question: Is it bad if a sample range falls *below* LCL_R?

A range below LCL_R signals an unexpected **drop in variability**—either a breakthrough improvement worth investigating and locking in, or a stuck/faulty measurement gauge!

In-Class Exercise

Wiper Blade Replacement Service Time

Service center wiper-blade replacement times monitored across $N = 6$ samples of size $\mathbf{n} = \mathbf{20}$ measurements (in minutes):

Sample	Mean ($\bar{X}$)	Range (R)
1	3.06	0.42
2	3.15	0.50
3	3.11	0.41
4	3.13	0.46
5	3.06	0.46
6	3.09	0.45

Questions

1. Determine the upper and lower limits for the $\bar{X}$ and R charts.
2. Is the process in control?

n	A_2	D_3	D_4
6	0.48	0.00	2.00
20	0.18	0.41	1.59

Solution: Wiper Blade Replacement Exercise

Step 1: Grand Mean ($\bar{\bar{X}}$), Average Range ($\bar{R}$), and Sample Size ($n = 20$)

$$\bar{\bar{X}} = \frac{3.06 + 3.15 + 3.11 + 3.13 + 3.06 + 3.09}{6} = \mathbf{3.10}\ \text{min}, \quad \bar{R} = \frac{0.42 + 0.50 + 0.41 + 0.46 + 0.46 + 0.45}{6} = \mathbf{0.45}\ \text{min}$$

Since each sample contains $\mathbf{n = 20}$ measurements, use $A_2 = 0.18$, $D_3 = 0.41$, and $D_4 = 1.59$.

$\bar{X}$-Chart Control Limits

$$(LCL_{\bar{X}},\ UCL_{\bar{X}}) = \bar{\bar{X}} \pm A_2\bar{R} = 3.10 \pm 0.18(0.45)$$

$$= (\mathbf{3.019},\ \mathbf{3.181})$$

All sample means $\in [3.06,\ 3.15]$ are inside!

R-Chart Control Limits

$$(LCL_R,\ UCL_R) = (D_3\bar{R},\ D_4\bar{R}) = 0.45 \times (0.41,\ 1.59)$$

$$= (\mathbf{0.185},\ \mathbf{0.716})$$

All sample ranges $\in [0.41,\ 0.50]$ are inside!

Conclusion: Yes, both charts are in control!

Process Capability

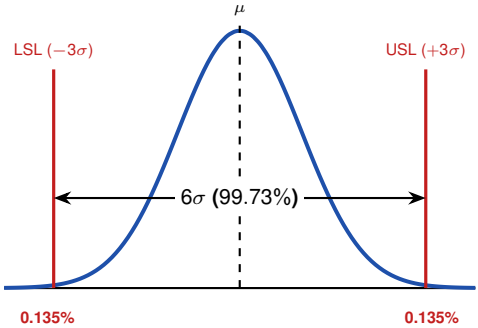
Meeting Customer Specification Limits (C_p and C_{pk})

A Common Measure of Process Capability (C_p)

Process Capability Ratio (C_p)

$$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma}$$

- USL = Upper Specification Limit
- LSL = Lower Specification Limit
- σ = Process standard deviation
- $\mathbf{C_p = 1}$: Borderline capable
  (USL $-$ LSL $= 6\sigma \implies$ 0.27% defect rate if centered)
- $\mathbf{C_p < 1}$: Process **not** capable
- $\mathbf{C_p > 1}$: Process capable



Are the Following Processes Capable?

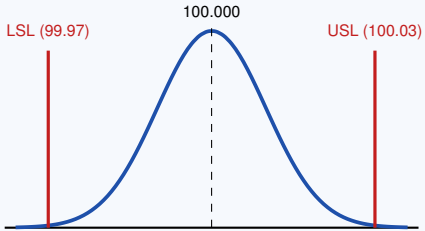
Process	Std. Dev. (σ)	USL	LSL	$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma}$	Capable? ($C_p \geq 1$)
A1	0.05	100.03	99.97		
A2	0.01	100.03	99.97		
B1	0.001	5.05	4.95		
B2	0.0005	5.05	4.95		
B3	0.02	5.05	4.95		

Are the Following Processes Capable? (Solution)

Process	Std. Dev. (σ)	USL	LSL	$C_p = \frac{\text{USL} - \text{LSL}}{6\sigma}$	Capable? ($C_p \geq 1$)
A1	0.05	100.03	99.97	**0.20**	**No**
A2	0.01	100.03	99.97	**1.00**	**Borderline ($C_p = 1$)**
B1	0.001	5.05	4.95	**16.67**	**Yes**
B2	0.0005	5.05	4.95	**33.33**	**Yes**
B3	0.02	5.05	4.95	**0.83**	**No**

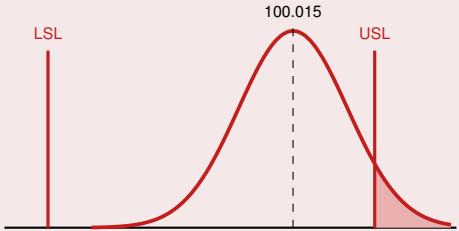
You Need to Be Careful With C_p (Centering Matters!)

$C_p = 1$ and Process Centered ($\mu = 100.000$)



- Below LSL (-3σ): 0.135%
- Above USL ($+3\sigma$): 0.135%
- **Total Defect Probability** $= 0.27\%$

$C_p = 1$ but Mean Shifted $+1.5\sigma$ ($\mu = 100.015$)



- Below LSL (-4.5σ): 0.00034%
- Above USL ($+1.5\sigma$): **6.68**%
- **Total Defect Probability** $= 6.68\%$**!**

Another Common Measure Is C_{pk} (Accounts for Centering)

Process Capability Index (C_{pk})

$$C_{pk} = \min\left\{\frac{\text{USL} - \mu}{3\sigma}, \frac{\mu - \text{LSL}}{3\sigma}\right\} \qquad \text{where } \mu = \text{process mean}$$

Process A2 Centered ($\mu = 100.000$)

Given USL $= 100.03$, LSL $= 99.97$, $\sigma = 0.01$:

$$C_{pk} = \min\left\{\frac{100.03 - 100.00}{3(0.01)}, \frac{100.00 - 99.97}{3(0.01)}\right\}$$
$$= \min\{1.0,\ 1.0\} = \mathbf{1.00}$$

Process A2 Shifted ($\mu = 100.015$)

Mean shifts up by 1.5σ to $\mu = 100.015$:

$$C_{pk} = \min\left\{\frac{100.03 - 100.015}{3(0.01)}, \frac{100.015 - 99.97}{3(0.01)}\right\}$$
$$= \min\{0.5,\ 1.5\} = \mathbf{0.50} \text{ (Not Capable!)}$$

Summary

1. Statistical Process Control (SPC)

- **Common vs. Assignable Causes:** Control charts separate natural random noise from actionable process shifts.
- **$\bar{X}$-Chart ($\bar{\bar{X}} \pm A_2\bar{R}$):** Tracks shifts in the **process mean**.
- **R-Chart ($D_3\bar{R},\ D_4\bar{R}$):** Tracks shifts in **process dispersion**. Both charts must be in control!

2. Process Capability (C_p & C_{pk})

- **In-Control $\neq$ Capable:** Control limits reflect voice of the *process*; specification limits (LSL, USL) reflect voice of the *customer*.
- **Potential Capability (C_p):** Measures whether $\text{USL} - \text{LSL} \geq 6\sigma$.
- **Actual Capability (C_{pk}):** Penalizes off-center processes by measuring distance from μ to the nearer specification limit.